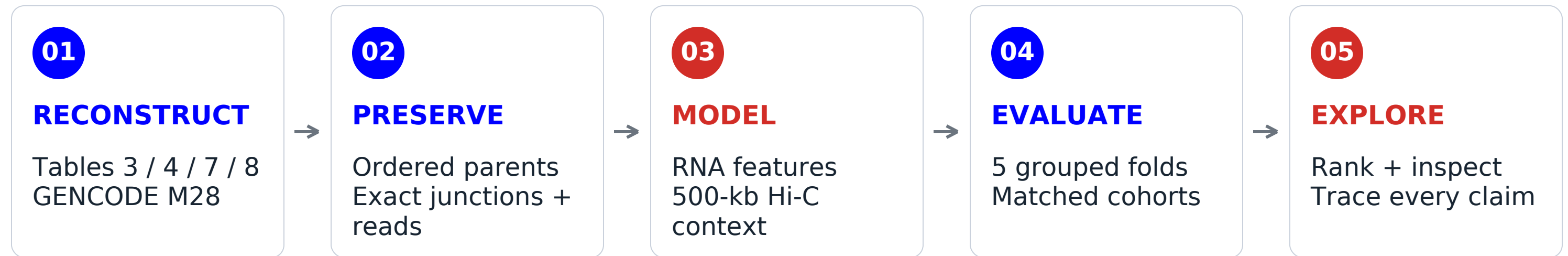


From published junctions to testable evidence

A reproducible workflow for mouse macrophage chimeric RNA candidates.



NVIDIA · independent evidence

Bounded Parabricks pilot → exact junction matching → display only

OpenAI · source-grounded explanation

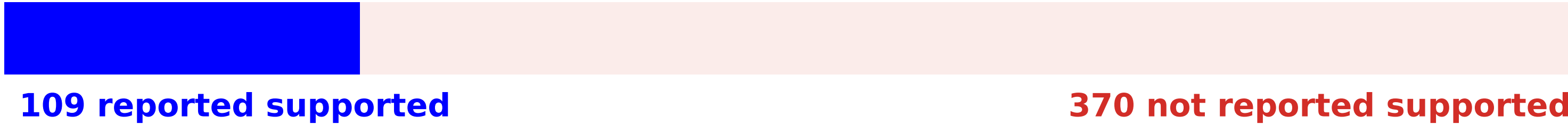
Evidence rows + paper passages → reviewed cached reports

The evaluation unit is an ordered gene pair

529 probe designs do not mean 529 independently validated chimeras.



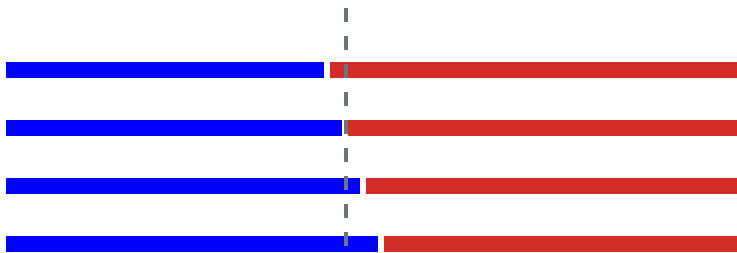
479 eligible pairs: one label per ordered pair



Different evidence answers different questions

Keep read identity, junction identity, pair-level reporting, and protein function separate.

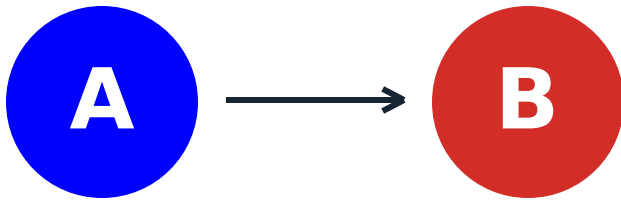
READ / JUNCTION



Exact coordinates + read IDs

What molecule or breakpoint was observed?

ORDERED PAIR



Reported $\neq$ every isoform validated

Was the pair reported
NanoString-supported?

FUNCTION



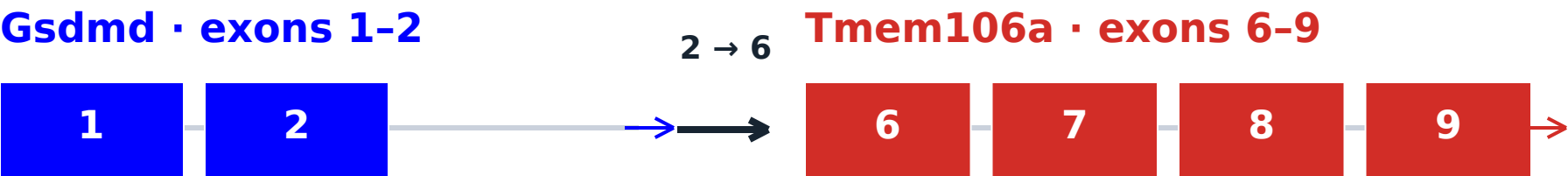
Requires separate experiments

Was a specific biological effect demonstrated?

Testing/QC unknown • Biological sample count unavailable • Missing contact $\neq$ zero contact

Gsdmd → Tmem106a: RNA architecture and a predicted fold

Reference-reconstructed RNA and protein sequence; the candidate fold remains low confidence.



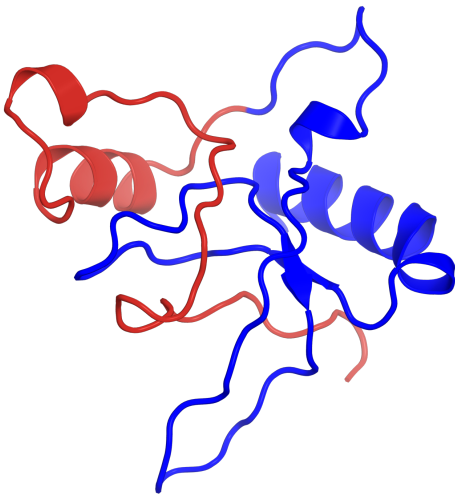
chr15:75,735,477 (+) → chr11:101,479,679 (+)
ENSMUST00000023238.6 / ENSMUST00000039581.14

118-aa reconstructed protein architecture



Blue: in-frame Gsdmd sequence. Red: novel out-of-frame Tmem106a-derived C terminus.

Candidate fold · predicted



Low confidence · mean pLDDT 48.70

Boltz2 · pTM 0.336

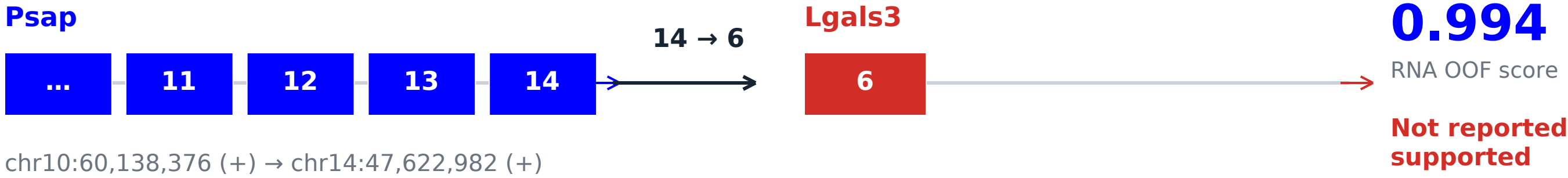
Junction evidence and model score can disagree

Exact exon choices are transcript-specific; a high score does not create biological support.

Cd274 → Lacc1



Psap → Lgals3

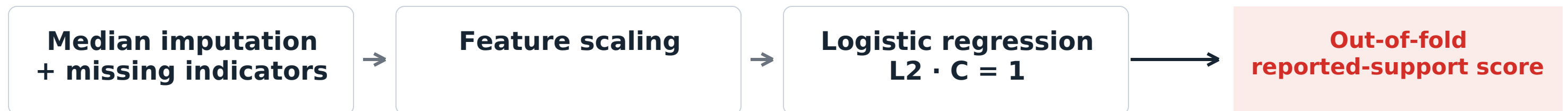


Evaluate new parent-gene groups, not memorized genes

352 parent-gene components are held intact across 5 folds.



Fit on training rows only

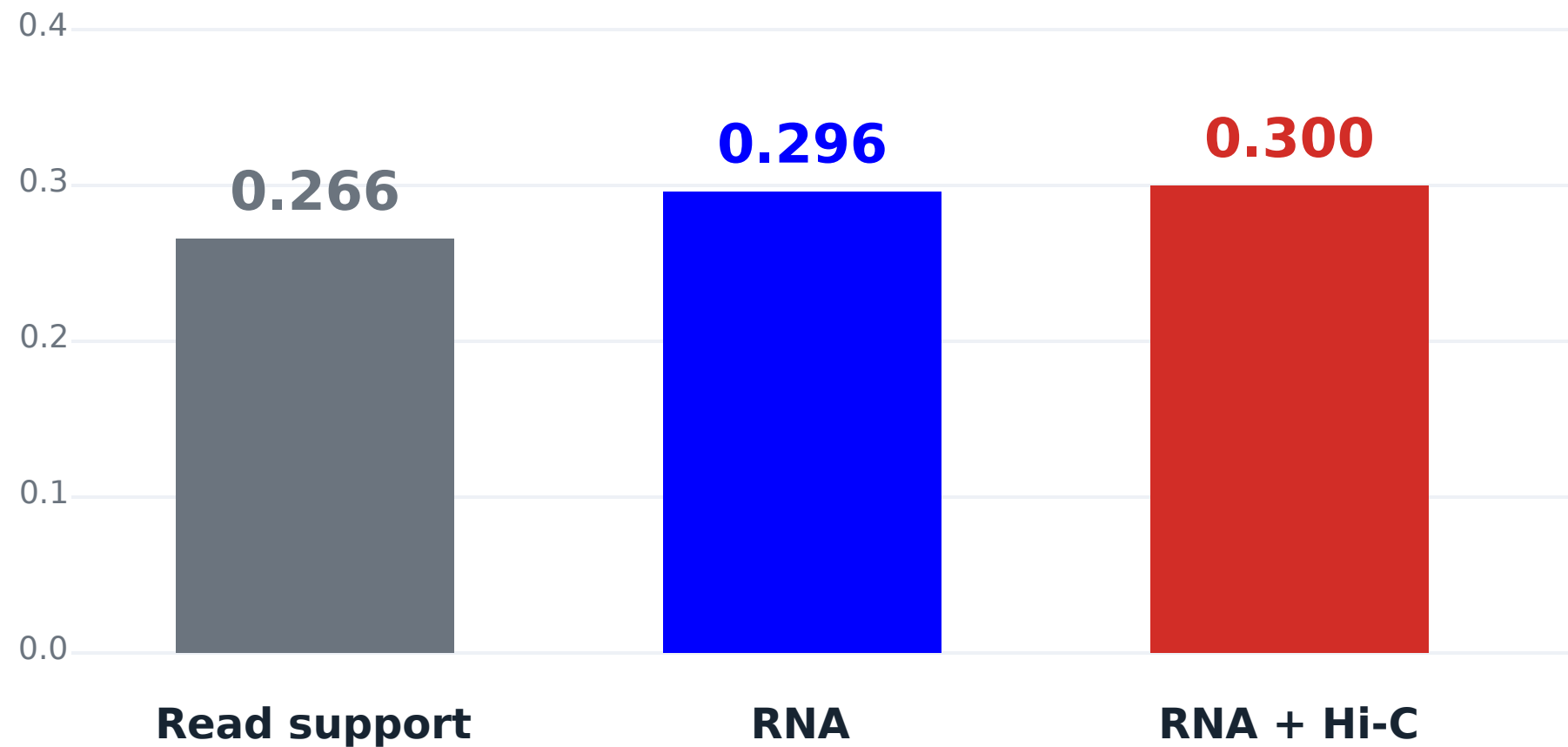


Score once

Hi-C ablation: both classifiers use identical contact-observed training and test candidates.

A small AP increase; uncertainty includes no benefit

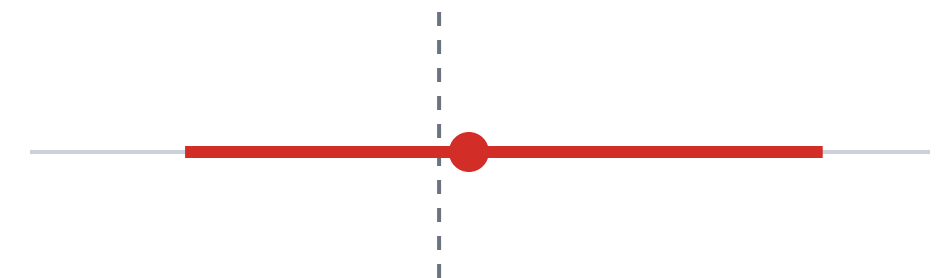
Matched cohort: 401 ordered pairs, 92 reported supported. Identical training rows and fixed folds.



Hi-C – RNA

+0.0036

difference in average precision



95% interval [-0.031, 0.047]

Top-ranked candidates tell a more cautious story

A small AP change does not guarantee better prioritization at a chosen cutoff.

Read-support
ranking



9 / 20

RNA



10 / 20

RNA + Hi-C



9 / 20

Interchromosomal matched subset: RNA 5 / 20 → RNA + Hi-C 7 / 20 (244 pairs; 58 reported supported)

An actual NVIDIA run, with no probe-panel matches

Parabricks on an A100-SXM4-80GB, using a deterministic two-million-pair RNA subsample.

VALIDATED READ PAIRS

2,000,000

Deterministic prefix subsample

ALIGNMENT WALL TIME

85.49 s

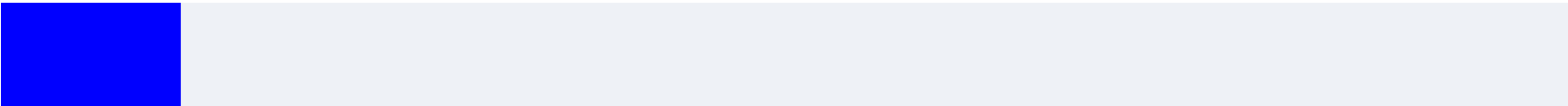
Reference / setup excluded

MATCHED PROBE PAIRS

0

Independent GPU evidence only

28,482 raw chimeric records



3,275 resolved split-junction records

25,207 encompassing-mate records



± 10 nt
ordered-junction match

\$1.35 quote-based instance estimate
Instance deleted; watchdog cancelled

OpenAI turns cached evidence into traceable explanations

Five Codex-authored reports separate observations, interpretations, and unknowns.



Source identity + exact input hash

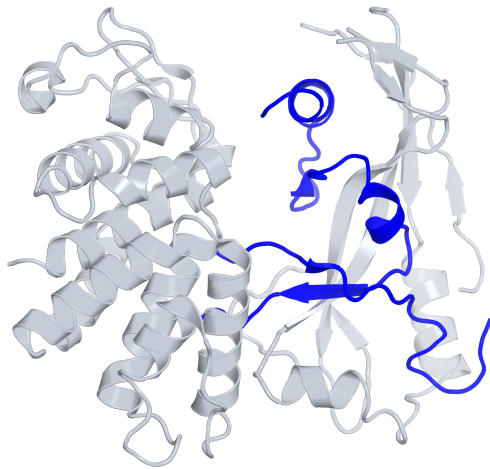
Supported: Gsdmd:Tmem106a, Cd274:Lacc1
Contrast: Psap:Lgals3 is not reported supported

Observed evidence stays distinct from biological authenticity, translation, and function.

A predicted chimera is not an experimental structure

Protein architecture can be source-grounded while the three-dimensional fold remains uncertain.

Experimental parent · GSDMD



PDB 6N9N · experimental mouse parent
3.30 Å crystallographic structure

Predicted candidate · 118 residues



Boltz2 candidate prediction · mean pLDDT 48.70
pTM 0.336 · low confidence

Blue residues 1-73: Gsdmd-derived. Red residues 74-118: novel out-of-frame C terminus.

An auditable result. An independent test comes next.

This gallery demonstrates an auditable experiment and a working evidence product.

Demonstrated

- 479 reproducible evaluated pairs
- Gene-disjoint held-out predictions
- Real GPU-generated junction output
- Five reviewed source-grounded reports

Still unestablished

- A consistent Hi-C ranking benefit
- Candidate recovery in the GPU pilot
- Assay QC / biological sample counts
- A reliable experimental chimera fold

Next test: complete assay/QC records + independent candidates + targeted junction confirmation.